

HISABDO AI

DAY 13 — CASH FLOW FORECASTING

Detailed Requirements, .NET/C# Implementation, Job Steps & Acceptance Checklist

1. Purpose

Day 13 adds Cash Flow Forecasting to HisabDo. It uses verified HisabDo history to estimate future income, expenses, savings and, only when a trusted opening balance exists, closing balance. The project roadmap specifies 3–6 months of history plus income, expenses and recurring-expense information as the basis for this estimate.

The calculation/forecasting engine is the source of numeric truth. The LLM is an explanation layer only and must not invent, alter or recalculate financial values.

2. Day 13 Objectives

- Finalize Cash Flow Forecasting requirements.
- Identify historical data needed for forecasting.
- Define input/output DTOs and API contract.
- Define an explainable MVP forecasting methodology.
- Forecast expected income, expense and saving.
- Forecast closing balance only when opening balance is trusted.
- Handle insufficient/missing history safely.
- Define confidence/data-quality and limitations.
- Validate forecasts against historical/sample data.
- Prepare .NET implementation, tests, documentation and acceptance checklist.

3. Scope / Non-Scope

In Scope	Not in MVP
3–6 month historical cash-flow analysis	Investment/market prediction
Income forecast	External bank prediction
Expense forecast	Guaranteed future income
Expected saving	Fraud decisions
Closing balance with trusted opening balance	LLM financial calculations
Recurring-expense signals	Confirmed subscriptions without evidence
AI explanation of verified forecast	Changing source transactions

4. Required Historical Data

Field	Priority	Purpose
UserId	Required	User isolation/authorization
TransactionId	Required	Unique transaction
TransactionType	Required	Income vs Expense
Amount	Required	Financial calculation
Date	Required	Time-series history
Category	Recommended	Expense behavior
Description/Note	Recommended	Recurring pattern support
Recurring expense data	Recommended	Improve forecast
Opening/current balance	Required for closing forecast	Closing balance

Budget	Optional	Future Day 14 integration
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Before production integration, map these concepts to the actual HisabDo .NET entities, DbContext, authentication and repository conventions. Do not invent database columns.

5. Forecast Period & History

- Preferred history: 3–6 complete months where available.
- Forecast horizon must be explicit; MVP target is the next month.
- Use comparable calendar periods.
- Fewer than 3 usable months should be marked Low/Limited rather than treated as high-confidence.
- Future iterations can add weekly/daily forecasting after monthly MVP validation.

6. MVP Forecasting Methodology

Start with a simple, explainable baseline. Aggregate monthly income and expenses, use recent historical averages as the baseline, and separately consider recurring-expense evidence when available. Omesha should validate whether a trend-based refinement improves accuracy enough to justify it.

Metric	MVP Rule
Expected Income	Average of usable historical monthly income values
Expected Expense	Average monthly expense over the selected history
Expected Saving	Expected Income – Expected Expense
Expected Closing Balance	Trusted Opening Balance + Expected Saving
Confidence	Rule-based data-quality indicator
Method	HistoricalBaseline

7. Calculation Rules

Calculation	Formula
Monthly Income	Sum valid Income transactions
Monthly Expense	Sum valid Expense transactions
Monthly Saving	Monthly Income – Monthly Expense
Expected Saving	Expected Income – Expected Expense
Expected Closing Balance	Opening Balance + Expected Income – Expected Expense

No division-by-zero, hidden adjustment or missing-value guessing is allowed. Numeric values must be produced by the backend/forecasting engine.

8. Confidence & Limitations

Data Condition	Suggested Confidence
6 usable months with reasonably complete history	High
3–5 usable months	Medium
<3 usable months	Low
No usable transactions	Unavailable
Major missing periods / inconsistent data	Limited
No trusted opening balance	Closing balance unavailable

- Confidence is a data-quality/forecast-reliability indicator, not a probability guarantee.
- Actual future results can differ from forecasts.
- Recurring candidates are signals, not guaranteed future transactions.
- The AI must explain limitations using only backend-provided facts.

9. Verified API Contract

Recommended endpoint: GET /api/ai/cash-flow-forecast/{userId}?months=6&horizonMonths=1

Example response:

```
{
  "userId": "USER_ID",
  "historyMonthsUsed": 6,
  "forecastHorizonMonths": 1,
  "expectedIncome": 150000.00,
  "expectedExpense": 108500.00,
  "expectedSaving": 41500.00,
  "openingBalance": 200000.00,
  "expectedClosingBalance": 241500.00,
  "confidence": "Medium",
  "method": "HistoricalBaseline",
  "limitations": [
    "Forecast is based on historical transaction behavior"
  ]
}
```

10. Architecture

HisabDo Database → Authenticated User Scope → Historical Data Service → Forecasting Engine → Verified Forecast Result → AI Explanation Layer → Mobile/Web/User

The same verified backend result should be reusable by Flutter and web. AI receives the verified result; it does not become the financial calculation engine.

11. Detailed Team Job Steps — Taha / Team Lead

1. Create/confirm the Day 13 feature branch and objective.
2. Review Day 10–12 architecture and preserve the verified-calculation/AI-explanation separation.
3. Confirm real HisabDo transaction, user and balance entities with Jaffer.
4. Confirm fields for income, expense, amount and date.
5. Confirm how opening/current balance is stored or calculated.
6. Finalize forecast horizon and history requirements.
7. Approve insufficient-data behavior.
8. Review and approve Omesha's MVP forecasting methodology.
9. Approve Jaffer's DTO/API contract.
10. Define Aarti's AI input as the verified forecast result only.
11. Coordinate Laiba's UI fields and limited/no-data states.
12. Review calculations with sample data.
13. Run holdout-style historical validation where possible.
14. Record errors, limitations and edge cases.
15. Approve documentation and implementation after acceptance checks pass.

12. Jaffer — Backend Job Steps

16. Identify the real HisabDo transaction entity/repository.
17. Implement authenticated user-scoped history retrieval.
18. Filter valid transactions.
19. Aggregate monthly income and expense.

20. Build reusable CashFlowForecastService.
21. Validate forecast input parameters.
22. Implement the historical baseline.
23. Calculate expected saving.
24. Calculate closing balance only when opening balance is trusted.
25. Return confidence and limitations metadata.
26. Add error/cancellation handling.
27. Add unit/integration tests.
28. Expose the forecast API after service validation.

13. Omesha — AI/ML Job Steps

29. Review available historical data and forecasting objective.
30. Define the MVP baseline/trend methodology.
31. Evaluate average-based and trend-based approaches.
32. Prepare historical holdout/sample datasets.
33. Compare predicted income/expense with actual values.
34. Calculate MAE/MAPE where mathematically appropriate.
35. Identify unstable and insufficient-data cases.
36. Document assumptions and limitations.
37. Recommend the final MVP approach based on validation.
38. Keep numeric forecasting outside the LLM.

14. Aarti — AI Explanation Job Steps

39. Define forecast explanation structure.
40. Accept only verified backend forecast values.
41. Explain expected income, expense and saving.
42. Explain closing balance only if supplied.
43. Explain confidence/data quality.
44. Explain forecast limitations.
45. Generate cautious actionable observations.
46. Never invent or recalculate amounts.
47. Prepare positive, negative and limited-data sample responses.
48. Test missing values and prompt-injection behavior.

15. Laiba — UI Job Steps

49. Create Cash Flow Forecast section/card.
50. Show forecast period.
51. Show expected income, expense and saving.
52. Show expected closing balance when available.
53. Show confidence/data-quality indicator.
54. Show AI explanation.
55. Clearly label all predicted values as forecast/estimated.
56. Create limited-data and no-data states.
57. Handle API loading/error states.
58. Test mobile/web presentation.

16. Testing & Edge Cases

- 3 months history.
- 6 months history.
- <3 months history.
- No transactions.
- Only income.
- Only expenses.
- Zero income.
- Zero expenses.
- Negative historical cash flow.
- Large one-time expense.
- Irregular income.
- Irregular expenses.
- Missing category/description.
- Invalid dates.
- Duplicate transactions.
- Missing historical month.
- No trusted opening balance.
- User A data isolation from User B.
- AI cannot modify backend forecast numbers.
- Prompt injection in transaction descriptions cannot override system rules.

17. Forecast Validation

Use a holdout-style check when possible: use earlier historical months to generate a forecast for a known later month, then compare predicted values with actual values.

Metric	Validation
Income	Predicted vs actual
Expense	Predicted vs actual
Saving	Predicted saving vs actual saving
Closing balance	Check only with reliable opening balance
Confidence	Correct classification for data quality
Security	User isolation
AI	No number invention/modification

18. Day 13 Acceptance Checklist

- ☐ Requirements finalized.
- ☐ Real HisabDo fields confirmed.
- ☐ 3–6 month behavior documented.
- ☐ Forecast horizon defined.
- ☐ MVP methodology approved.
- ☐ Income forecast implemented/tested.
- ☐ Expense forecast implemented/tested.
- ☐ Saving forecast implemented/tested.
- ☐ Closing balance handled safely.
- ☐ Insufficient-data handling implemented.
- ☐ Confidence/limitations documented.
- ☐ Verified API contract defined.
- ☐ AI receives verified result only.

- ☐ UI labels forecasts clearly.
- ☐ Historical/sample validation completed.
- ☐ Edge cases tested.
- ☐ Authorization/data isolation tested.
- ☐ Documentation completed.

19. Final Definition

HisabDo AI Cash Flow Forecasting is an explainable forecasting module that uses verified historical financial data—preferably 3–6 months of income, expenses and recurring-expense signals—to estimate future income, expenses and savings, and to estimate closing balance when a trusted opening balance is available. The forecasting engine produces the numbers; AI explains the verified result and its limitations.

20. Implementation Warning

This document is an MVP implementation guide. It is not a claim about the exact current HisabDo production schema. Before merging, map the reference DTOs/repository to the real .NET entities, DbContext, authorization and existing API conventions.